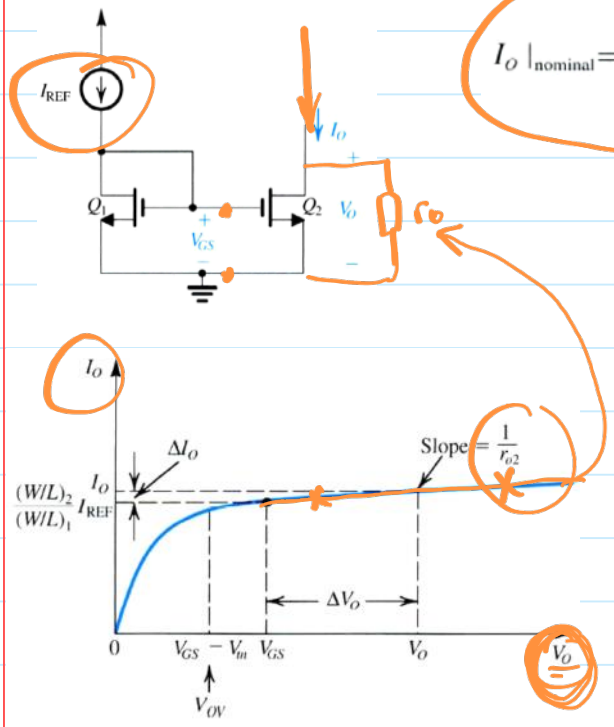
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 8.2 IC Biasing: Current Sources and Current Mirrors

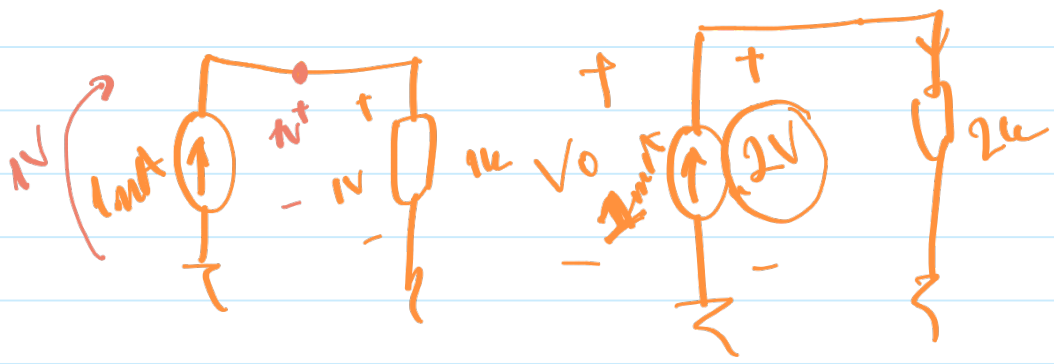
Recap



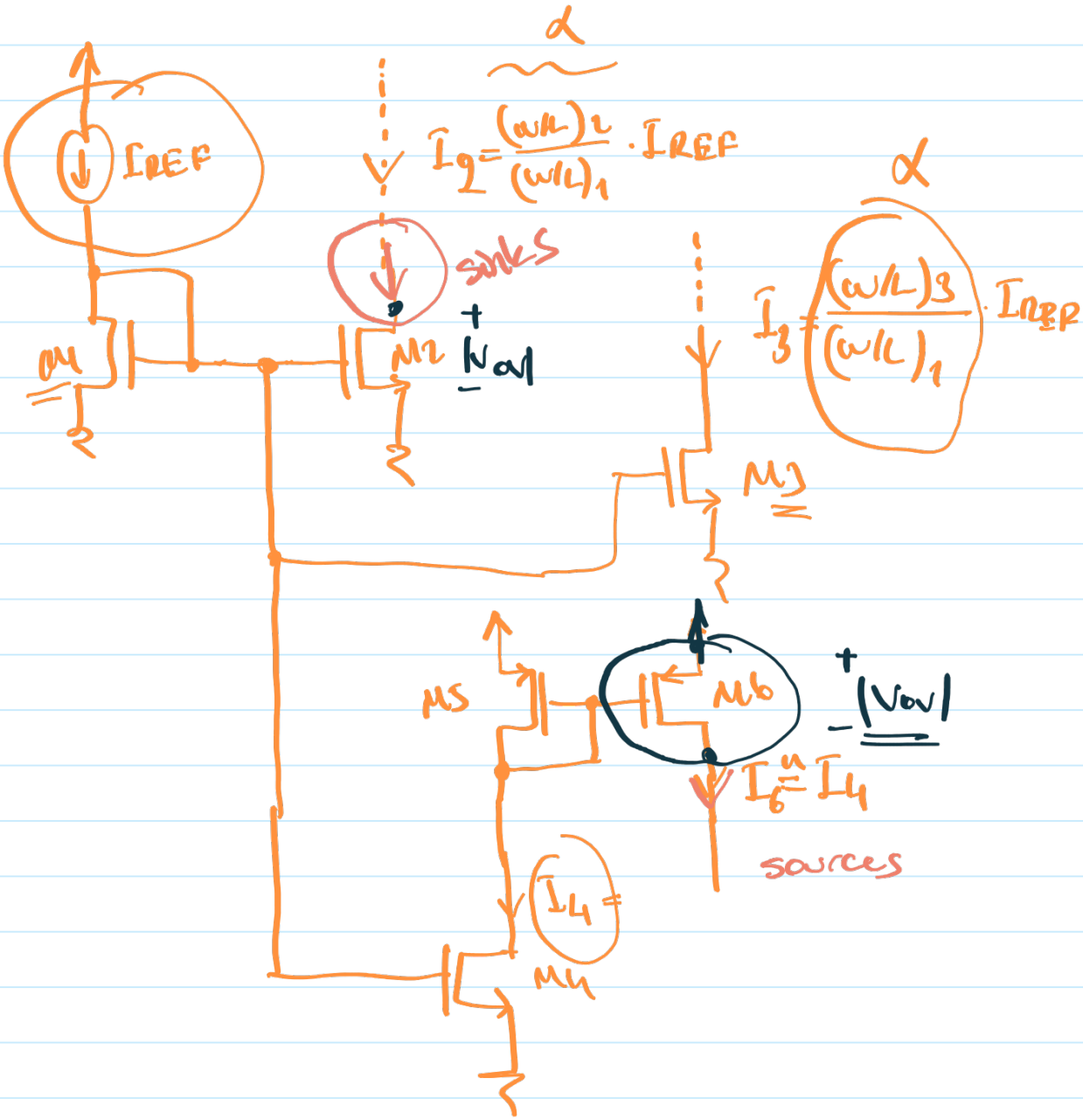
$$I_O|_{\text{nominal}} = \frac{(W/L)_2}{(W/L)_1} I_{REF} \qquad V_O \geq V_{OV}$$

0.0001 —

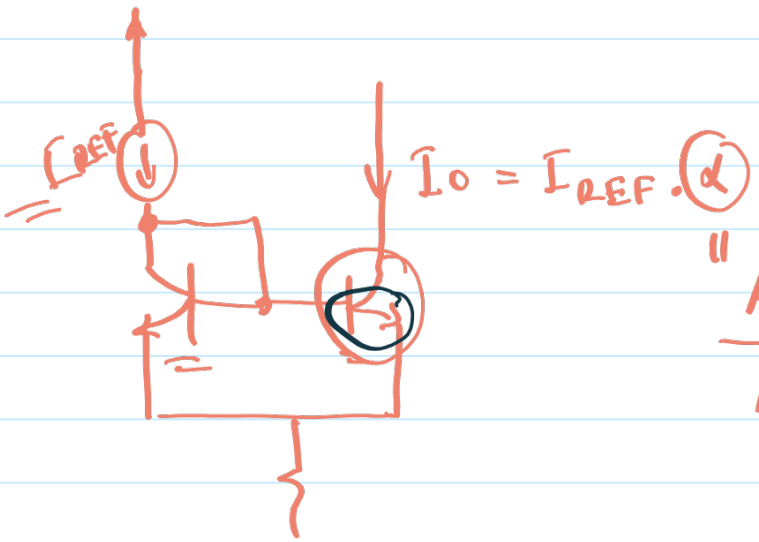
$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF}\left[1 + \frac{V_O - V_{GS}}{V_{A2}}\right]$$

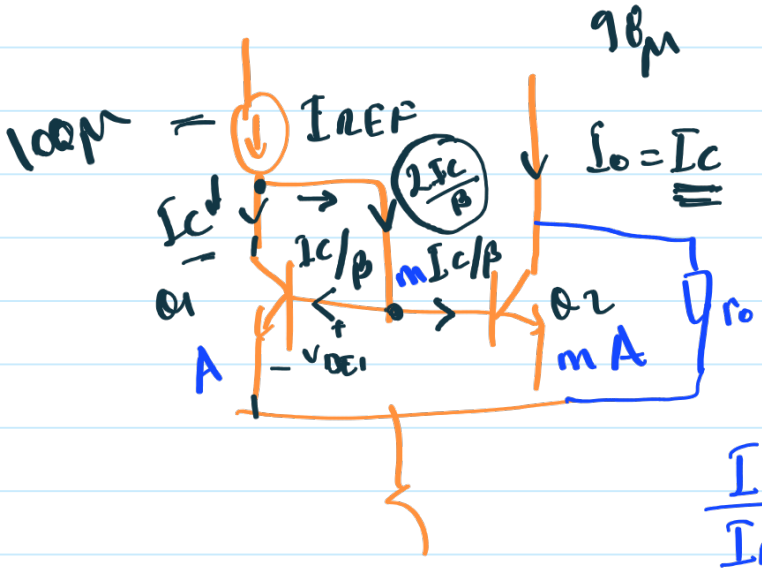


$I_o = I_{REF} \cdot \alpha$



BJT Current Mirror



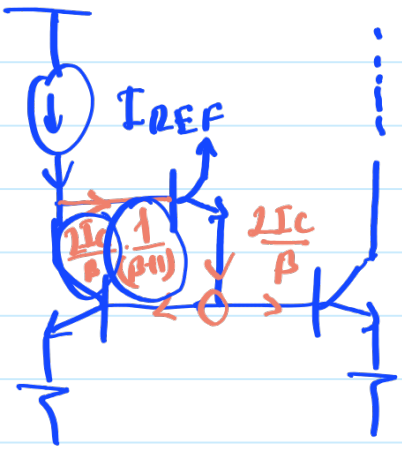


$$I_{REF} = I_C + \frac{2I_C}{\beta}$$

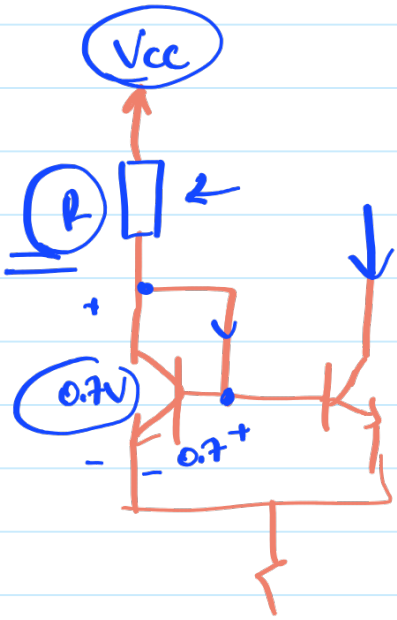
$$\frac{I_O}{I_{REF}} = \frac{I_C}{I_C\left(1+\frac{2}{\beta}\right)} = \frac{1}{1+\frac{2}{\beta}}$$

$$\frac{I_O}{I_{REF}} = \frac{m}{1+\frac{m+1}{\beta}}$$

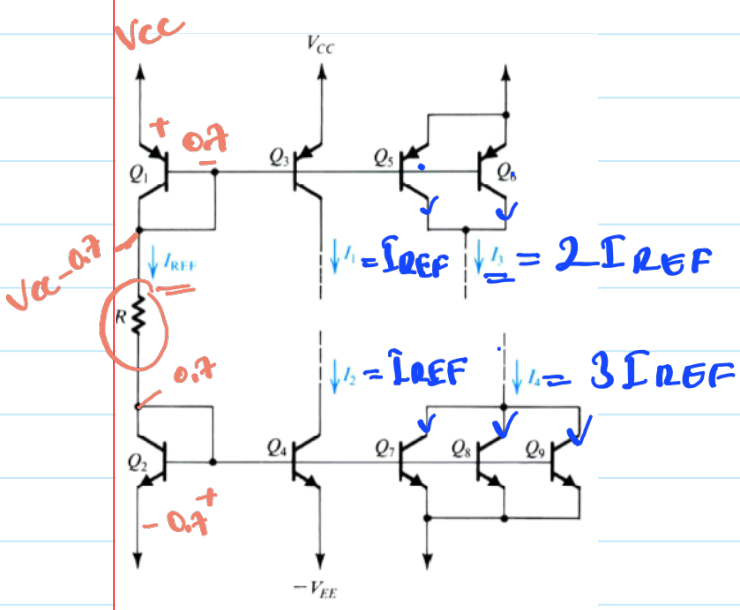
$$I_O = I_{REF}\,\frac{m}{1+\frac{m+1}{\beta}}\left[1+\frac{V_O - V_{OE}}{V_{A2}}\right]$$



BJT Current Source and Steering



$$\frac{V_{CC} - 0.7}{R} = I_{REF}$$



$$\frac{V_{CC} - 1.4}{R} = I_{REF}$$

Example

Find the voltages at all nodes and the currents through all branches in the circuit of Fig. VE8.2. Assume $|V_{BE}| = 0.7$ V and $\beta = \infty$.

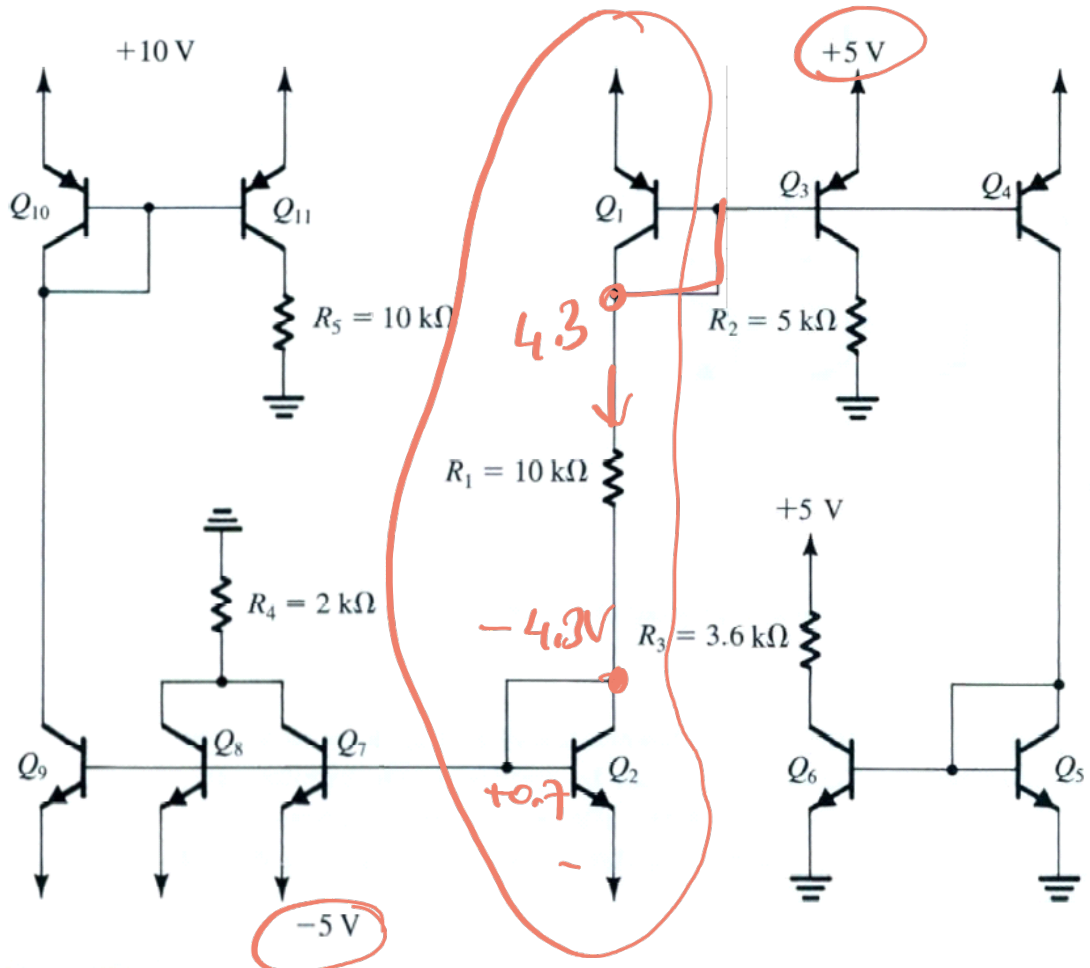
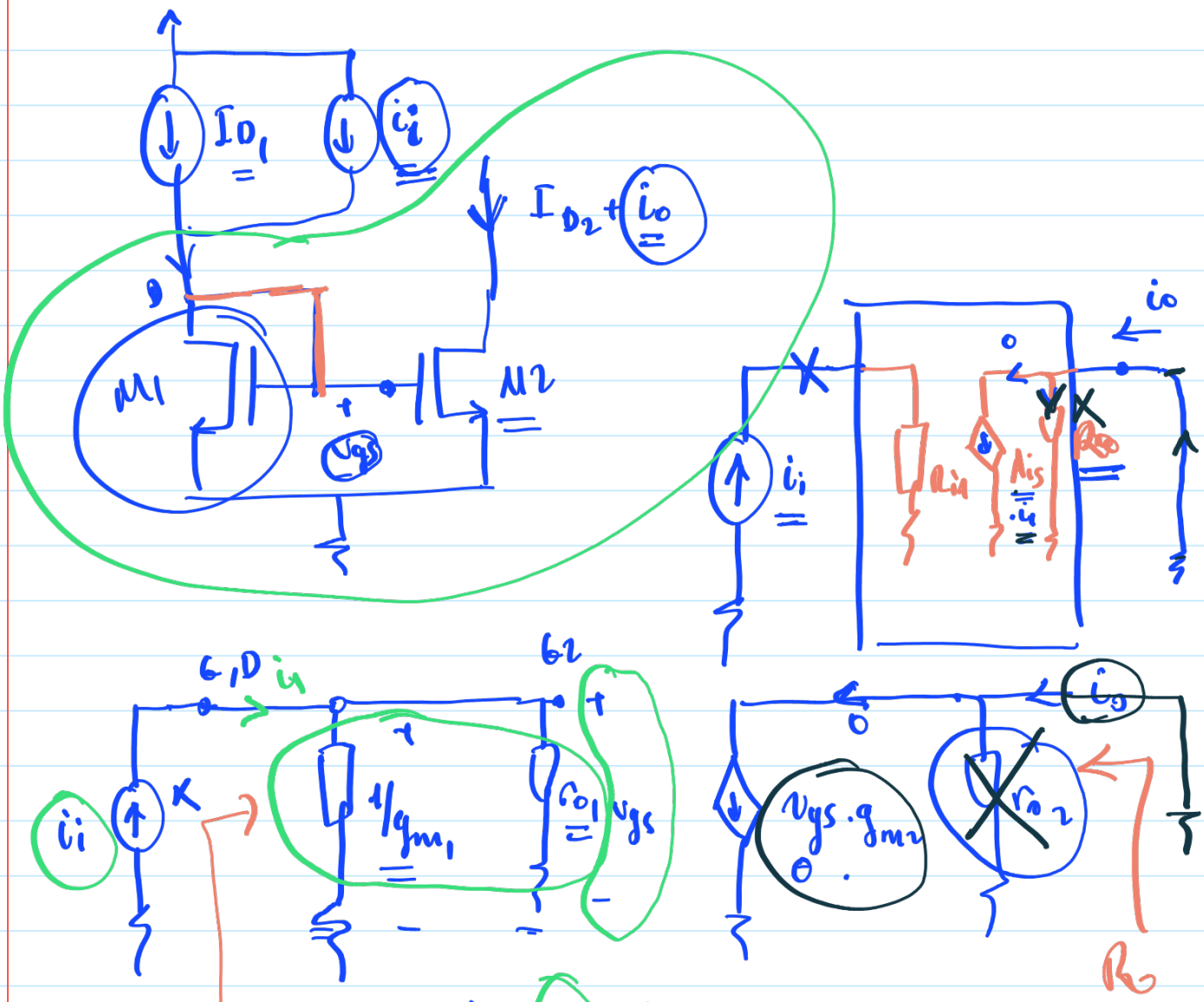


Figure VE8.2 Circuit for Video Example 8.2.

Base Current Compensation

Small Signal Operation of Current Mirrors



$$R_M = r_{o_1} \parallel 1/g_{m_1} \cong \frac{1}{g_{m_2}} \qquad A_{is}$$

$$v_{gs} = (r_{o_1} \parallel 1/g_{m_1})\, i_i$$

$$R_{o_2} = r_{o_2}$$

$$A_{is} = \left.\frac{i_o}{i_i}\right|_{V_o=0} = \frac{g_{m_2} v_{gs}}{i_i} = \frac{g_{m_2}}{g_{m_1}} \cdot \frac{1}{1 + 1/g_{m_1} r_{o_1}}$$

$$\mu_n C_{ox}\, W/L\, V_{ov}$$

$$n = \frac{(W/L)_2}{(W/L)_1}\left[\frac{1}{1 + 1/g_{m_1} r_{o_1}}\right]$$

Template